

Logistic Regression as ML

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Introduction

Logistic regression models class probability as $P(y = 1 \mid x) = \sigma(w^\top x + b)$, where σ is the logistic function, and fits parameters by minimising the cross-entropy loss. As an ML workhorse it is well-calibrated by construction, interpretable through coefficients on the log-odds scale, regularisable with L1, L2, or elastic-net penalties, and scales to millions of features — making it the standard baseline for text classification, click-through prediction, and medical risk scoring.

Prerequisites

A working understanding of linear models, the cross-entropy / log-loss objective, and regularisation (L1 vs L2 penalties).

Theory

The regularised log-loss objective is

$$L(w) = - \sum_i [y_i \log \sigma(w^\top x_i) + (1 - y_i) \log(1 - \sigma(w^\top x_i))] + \lambda R(w),$$

with $R(w) = \|w\|_2^2$ for L2 (ridge), $\|w\|_1$ for L1 (lasso), or a convex combination for elastic net. Optimisation uses Newton-Raphson (IRLS) for the unregularised case and coordinate descent (`glmnet`) for the L1/elastic-net case. The penalty parameter λ is tuned by cross-validation, with `lambda.min` minimising CV deviance and `lambda.1se` providing a more parsimonious choice within one SE of the minimum.

Assumptions

Log-odds are linear in the predictors; observations are independent; sufficient sample size per feature (rule of thumb: $n > 10p$ for unregularised, less for regularised models).

R Implementation

```
library(glmnet)

set.seed(2026)
n <- 400; p <- 30
X <- matrix(rnorm(n * p), n, p)
beta <- c(rep(1, 5), rep(0, 25))
y <- rbinom(n, 1, plogis(X %*% beta - 1))

# Cross-validated lasso
cv_fit <- cv.glmnet(X, y, family = "binomial",
                    alpha = 1, nfolds = 10)
plot(cv_fit)
```

```
coef_min <- as.vector(coef(cv_fit, s = "lambda.min"))
head(coef_min, 10)
cat("# nonzero:", sum(coef_min != 0) - 1, "\n")
```

Output & Results

`cv.glmnet()` returns the cross-validation deviance curve over λ , the recommended `lambda.min` and `lambda.1se`, and the fitted coefficient paths. Lasso correctly recovered the 5 non-zero coefficients in the data-generating model; the deviance plot shows the U-shaped CV-error curve.

Interpretation

A reporting sentence: “Lasso logistic regression with cross-validated $\lambda_{\min} = 0.03$ retained 5 non-zero features that exactly match the data-generating β ; AUC on a held-out test set was 0.84 with calibration intercept -0.02 and slope 0.97 — well-calibrated. The sparser `lambda.1se` solution retained 3 features with AUC 0.79.” Always report the regularisation choice and a calibration check.

Practical Tips

- Standardise features before regularisation; `glmnet` does this internally and back-transforms, but explicit pre-scaling makes the workflow more transparent.
- Use `lambda.1se` for a sparser, more interpretable model; `lambda.min` is typically more accurate but less parsimonious.
- Logistic regression produces well-calibrated probabilities for correctly-specified models; SVMs, trees, and neural networks typically need Platt scaling or isotonic regression to recover calibration.
- For massively sparse data (text bags-of-words, click logs), `LiblineaR` is dramatically faster than `glmnet`; it handles millions of features routinely.
- For multi-class problems, `family = "multinomial"` in `glmnet` extends the framework natively; alternatively, `nnet::multinom()` for unregularised multinomial regression.
- For very imbalanced classes, combine logistic regression with class weights (`weights = ...`) or threshold tuning on a validation set.

R Packages Used

`glmnet::cv.glmnet()` for ridge, lasso, and elastic-net logistic regression; `LiblineaR` for fast sparse linear classifiers; base R `glm(family = binomial)` for unregularised logistic regression; `parsnip::logistic_reg()` for tidymodels-flavoured wrappers; `pROC` for ROC and calibration diagnostics.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Cross-validation, nested CV, bootstrap .632+
- Regularisation: ridge, lasso, elastic net
- Trees, random forests, gradient boosting
- Interpretability and SHAP
- Tabular neural networks with torch

- Imaging and sequence models intro
- tidymodels pipelines
- FDR, knockoffs, replication crisis